# EPSR-test-ex11 Additional Gaussian pair potentials

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

## What this case tests

Exercises the additional pair potentials of the `epsr` engine [1,2]: Gaussian terms h·exp(-(r - r0)²/(2w²)) appended to a `.pcof` pair line are added to the reference potential, letting users impose known features such as hydrogen-bond wells. A negative position also excludes the pair from EP refinement [3].

## Code exercised

**Fortran (original, fortran_src/):**

- `read_and_save_files.f`: `read_pcof` (`(r, w, h)` triples after `rminex pcoefex coordex`; negative r or w sets `efacp` = 0), `save_files_others` (`(3(1x,f10.5),N(1x,f9.4,1x,f9.4,1x,e11.4))` format)
- `epsr_standalone_rev3.f`: `generate_rp` (additional terms added to the reference potential), `ptgnsq` (EP update scaled by `efacp`), `generate_ep`

**Rust (port, rust/epsr-core/src/):**

- `pcof.rs`: `read_pcof`, `write_pcof_pairs`
- `potential.rs`: `generate_rp`, `ptgnsq`, `generate_ep`

## Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. Gaussian terms are added on three pairs: S-O1 (two terms: r 3.5 Å, w 0.2 Å, h 0.3 and r 4.5 Å, w 0.3 Å, h -0.2), C-C (r -3.0, w 0.3, h 0.1) and water O-H (r 1.8, w 0.15, h -0.5). The negative position on C-C sets that pair's EP factor `efacp` to 0, so its EP is switched off and not refined, while the term itself is applied at |r| = 3.0 Å.

## Inputs

Copy of EPSR-test with only three `.pcof` pair lines changed (`S O1`, `C C`, `O1 H1`): `0 0 0` followed by the triples above.

## How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
Two chained passes, so the Gaussian triples are written in the special format and read back. In the Fortran reference the C-C entry keeps its triple (written as `-3.0000 0.3000 0.1000E+00`) and its leading EP coefficient is 0, confirming `efacp` = 0. All 27 output files are identical after both passes.

## References


[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.